

Answers learning task 1

2026 – 2027

In Questions 2.1, 2.8 and 2.17, provide a reasonable description of the sample space for each of the random experiments. There can be more than one acceptable interpretation of each experiment. Describe any assumptions you make.

Exercise 2.1: Each of three machined parts is classified as either above or below the target specification for the part.

Answer

$S = \{aaa, aab, aba, baa, abb, bab, bba, bbb\}$, where a = above and b = below

Exercise 2.8: The following two questions appear on an employee survey questionnaire. Each answer is chosen from the five-point scale 1 (never), 2, 3, 4, 5 (always).

- Is the corporation willing to listen to and fairly evaluate new ideas?
- How often are my coworkers important in my overall job performance?

Answer

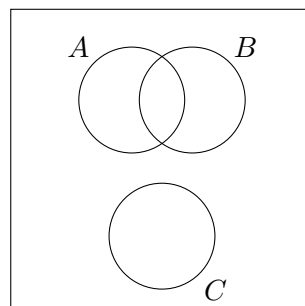
$S = \{(1, 1), (1, 2), \dots, (1, 5), (2, 1), \dots, (5, 5)\}; n_S = 25$

Exercise 2.17: Calls are repeatedly placed to a busy phone line until a connection is achieved.

Answer

$S = \{c, bc, bbc, bbbc, \dots\}$, where b = busy and c = connect

Exercise 2.20: Three events are shown on the Venn diagram in the following figure:



Reproduce the figure and shade the region that corresponds to each of the following events.

- | | |
|--------------------------|-----------------------------------|
| (a) A' | (b) $(A \cap B) \cup (A \cap B')$ |
| (c) $(A \cap B) \cup C$ | (d) $A \cap (B \cup C)'$ |
| (e) $(A \cap B)' \cup C$ | |

Answer

- (a) The region outside circle A .
- (b) Simplifies to A ; the entire circle A is shaded.
- (c) The intersection region of A and B , plus the entire circle C .
- (d) The portion of circle A that does not overlap with either B or C .
- (e) Everything outside the intersection of A and B , plus the entire circle C .

Exercise 2.29: The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. Define the events A and B as follows: $A = \{x \mid x < 65\}$ and $B = \{x \mid x > 45.5\}$.

Describe each of the following events:

- (a) A'
- (b) B'
- (c) $A \cap B$
- (d) $A \cup B$

Answer

- a. $[65, \infty)$ b. $(0, 45.5]$ c. $(45.5, 65)$ d. $(, \infty)$

Exercise 2.31: A sample of two printed circuit boards is selected without replacement from a batch. Describe the (ordered) sample space for each of the following batches:

- (a) The batch contains 90 boards that are not defective, seven boards with minor defects, and three boards with major defects.
- (b) The batch contains 90 boards that are not defective, eight boards with minor defects, and one board with major defects.

Answer

- a. $S = \{gg, gm, gb, mg, mm, mb, bg, bm, bb\}$,
where g = good, m = moderate and b = bad
- b. $\{gg, gm, gb, mg, mm, mb, bg, bm\}$

Exercise 2.55: The sample space of a random experiment is $\{a, b, c, d, e\}$ with probabilities 0.2, 0.1, 0.2, 0.3, and 0.2, respectively. Let A denote the event $\{a, b, c\}$, and let B denote the event $\{c, d, e\}$. Determine the following:

- (a) $P(A)$
- (b) $P(B)$
- (c) $P(A')$
- (d) $P(A \cup B)$
- (e) $P(A \cap B)$

Answer

- (a) 0.5
- (b) 0.7
- (c) 0.5
- (d) 1
- (e) 0.2

Exercise 2.74: If $P(A) = 0.4$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

- (a) $P(A')$
- (b) $P(A \cup B)$
- (c) $P(A' \cap B)$
- (d) $P(A \cap B')$
- (e) $P[(A \cup B)']$
- (f) $P(A' \cup B)$

Answer

- | | | |
|---------|---------|---------|
| (a) 0.6 | (b) 0.5 | (c) 0.1 |
| (d) 0.3 | (e) 0.5 | (f) 0.7 |

Exercise 2.77: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	10
	Low	15	5

- (a) If a disk is selected at random, what is the probability that its scratch resistance is high and its shock resistance is high?
- (b) If a disk is selected at random, what is the probability that its scratch resistance is high or its shock resistance is high?
- (c) Consider the event that a disk has high scratch resistance and the event that a disk has high shock resistance. Are these two events mutually exclusive?

Answer

- (a) 0.7
- (b) 0.95
- (c) No, because $P(A \cap B) = 0.70$. So $A \cap B \neq \emptyset$

Exercise 2.87: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	10
	Low	15	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Determine the following probabilities:

- | | |
|----------------|----------------|
| (a) $P(A)$ | (b) $P(B)$ |
| (c) $P(A B)$ | (d) $P(B A)$ |

Answer

- | | |
|-----------|------------|
| (a) 0.85 | (b) 0.8 |
| (c) 0.875 | (d) 0.8235 |

Exercise 2.91: The following table summarizes the analysis of samples of galvanized steel for coating weight and surface roughness:

		Coating Weight	
Surface Roughness	High	36	16
	Low	114	34

- (a) If the coating weight of a sample is high, what is the probability that the surface roughness is high?
- (b) If the surface roughness of a sample is high, what is the probability that the coating weight is high?
- (c) If the surface roughness of a sample is low, what is the probability that the coating weight is low?

Answer

- (a) 0.24
- (b) 0.6923
- (c) 0.2297

Exercise 2.106: Suppose that $P(A | B) = 0.25$, $P(A | B') = 0.35$, and $P(B) = 0.8$. What is $P(A)$?

Answer

0.27

Exercise 2.113: A batch of 30 injection-molded parts contains 6 parts that have suffered excessive shrinkage.

- (a) If two parts are selected at random, and without replacement, what is the probability that the second part selected is one with excessive shrinkage?
- (b) If three parts are selected at random, and without replacement, what is the probability that the third part selected is one with excessive shrinkage?

Answer

- (a) 0.2
- (b) 0.2